

Event-Driven Lighting for Immersive Attention Guidance

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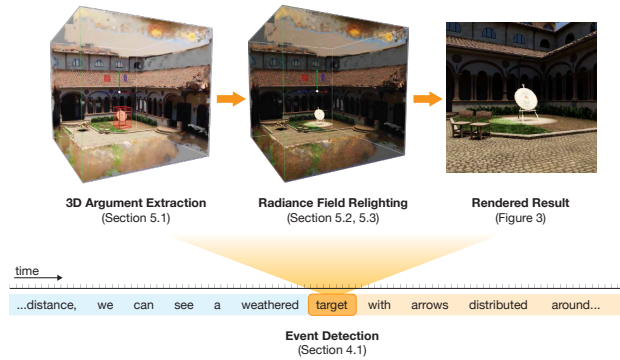


Figure 1: Visual overview of Event-Driven Lighting pipeline.

ABSTRACT

In immersive VR environments, users often miss events occurring outside their field of view. To remedy this, we propose a framework to guide user attention by dynamically adjusting environment lighting based on an input text script and 3D scene. Validated via user study, our system improves users' recall of virtual environments.

Index Terms: Attention Guidance, Event Extraction, Multimodal Systems, Radiance Fields, Cinematic Lighting.

1 INTRODUCTION

This work introduces *Event-Driven Lighting (EDL)*, a novel system that leverages dynamic lighting to guide user attention. Our system first detects key events within a provided input text script using large language models (LLMs), then locates corresponding objects within the input 3D scene, and finally automatically adjusts lighting to highlight those objects at the appropriate time.

2 RELATED WORK

Attention Guidance in Virtual Environments. In the virtual reality (VR) community, attention guidance methods have been pivotal across various domains (e.g. films and games), with the aim to direct users' focus towards their intended targets [13]. Various techniques have been proposed, including forced rotation [10], visual cues such as selective blurs [15] or arrows [7], and physical chair reorientation [4]. While effective, some methods can negatively impact immersion.

Event Extraction. The definition of an event has yet to be universally agreed upon [8]. One widely-used definition is "a specific occurrence, involving participants" [2], which we adopt, extending "participants" to mean general 3D objects. *Event Extraction* is the process of creating structured data from unstructured text. It is well

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studied in Information Extraction [3], and typically involves two phases: (1) detection and (2) argument extraction. We use LLMs [11] for zero-shot event detection.

Radiance Field Relighting. Existing radiance field methods [9] struggle with realistic relighting. Recent works try to disentangle scene lighting, geometry, and material [1]. We use an underlying radiance field with additive relighting based on DMRF [12].

3 EVENT-DRIVEN LIGHTING

3.1 Event Detection

Event Detection on a Script. Event detection begins by identifying subjects and their associated sentences in the script using a ChatGPT prompt. This process involves filtering subjects against a list of scene entities and mapping them to a set of valid event IDs.

Argument Extraction from Generated Audio. Given partially populated event schema from the event-detection stage, and the script itself, we can populate the start- and end-time arguments of our event by generating a narration of the script. The narration provides timing data, which is used to convert the start and end character ranges into timing values.

3.2 Event-Driven Scene Relighting

Argument extraction from 3D scenes. To extract 3D bounding boxes for events, we leverage LERF [6], which produces volumetric relevancy maps in a NeRF volume for a given language query, by embedding CLIP features into the NeRF. We extend LERF by extracting query bounding boxes.

Lighting Design. We use a simple lighting design recipe that applies to a wide range of scenarios, by simply placing a spot light as key directly above the subject bounding box. Note that lighting design can be a subjective art form; more sophisticated recipes for lighting design may be similarly effective at guiding user attention.

Radiance Field Relighting. This work leverages DMRF [12] to enable relighting of radiance fields. DMRF integrates traditional path tracing with NeRF rendering: at ray-surface intersections, path tracing determines illumination from light sources (point, spot, area). This illumination is then multiplied with the NeRF color to obtain the final pixel color.

4 EVALUATION

4.1 Qualitative Visual Results

We visualize event-driven relighting results in two scenes, *Classroom* and *Courtyard* (Figure 2) Relighting results looks physically plausible, and accentuate important events within the environment.

4.2 User Study

Experiment Design and Setup To assess the efficacy of our event-driven lighting (EDL) approach, we conducted a between-subject study involving two groups: one experiencing the scene with our EDL approach, and the other viewing the original scene without any lighting effects (denoted as *baseline*). Participants' experiences were evaluated post-trial using the Simulator Sickness Questionnaire (SSQ) [5] and Immersive Presence Questionnaire



Figure 2: **Visual comparison between default lighting and Event-Driven Lighting (EDL) in a NeRF environment.** Quoted entities are the detected arguments of the current event in the narration.

(IPQ) [14]. Additionally, we investigated the impact of our approach on users' ability to remember the virtual scene's content by designing two tasks: an information recall task requiring participants to identify objects present in the scene from a list of items, and a spatial memory task requiring them to circle the placement of objects mentioned by the narrator on a map of the virtual scene.

Two scene types were examined: a *static scene* generated using our NeRF-based EDL implementation (exported to a 360° video to ensure that participants were able to view the scene at a high frame rate) and a *dynamic scene* featuring animated objects implemented in the Unity game engine. For each scene, the narrator delivered a 90-second script describing nine objects distributed strategically around the participant (i.e., not all objects were placed within the user's field of view upon starting the experiment).

Participants. Twenty-two participants (15 males, 7 females) were recruited from university students (age range 19-33, $\mu = 27$). Participants self-reported their gaming experience as one of four categories: 11 non-gamers, 8 casual gamers, 2 core gamers, and 1 hardcore gamer. Four had no prior VR experience. Participants were evenly divided into two groups for the between-subject study.

Procedure. Participants, equipped with a Meta Quest Pro headset connected to a computer, were seated on a swivel chair for the duration of the study. Participants were immersed in a 360° scene. A training session familiarized them with viewing a 360°, requiring identification of virtual objects. Subsequently, participants completed two trials, viewing either the *static scene* or the *dynamic scene*, with the order of presentation counterbalanced.

Post-experiment, a semi-structured interview was conducted to elicit comments and feedback from participants. The entirety of the experimental process, from the introduction to the last interview, was completed within 30 minutes. Participants were compensated with a \$10 Amazon gift card upon completion of the experiment.

4.3 Results

To assess the normality of our data, we initially conducted a Shapiro-Wilk test, revealing a departure from normal distribution ($p < .001$). A Mann-Whitney U Test was employed for data analysis. Note that in the *Information Recall* and *Immersion* tasks, we did not note any improvement that was statistically significant.

Spatial Memory For the scene understanding task, participants labeled objects mentioned during the trial on a floor map based on their memory. *EDL* achieved a higher score ($\mu = 8$,

$\sigma = 1.15$) than the baseline ($\mu = 7.14$, $\sigma = 1.32$), indicating a significant difference ($Z = -2.205$, $p < 0.05$). This suggests that *EDL* effectively supports users in spatial memory tasks.

5 CONCLUSION

This study introduces the conceptual framework of Event-Driven Lighting (*EDL*), and demonstrates its effectiveness in improving users' spatial memory in VEs. Participants showed significantly better spatial memory with *EDL*, finding it helpful for navigating the narrative. Our results suggest that *EDL* can improve user experience in dynamic, interactive scenarios. Future directions include integrating *EDL* with user actions, exploring more complex lighting schemes, refining event detection and relighting techniques.

REFERENCES

- [1] M. Boss, V. Jampani, R. Braun, C. Liu, J. T. Barron, and H. P. Lensch. Neural-pil: Neural pre-integrated lighting for reflectance decomposition. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 1
- [2] J. M. K. M. Christopher Walker, Stephanie Strassel. Ace 2005 multilingual training corpus ldc2006t06, 2006. 1
- [3] S. Duan, R. He, and W. Zhao. Exploiting document level information to improve event detection via recurrent neural networks. In G. Kondrak and T. Watanabe, eds., *Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 352–361. Asian Federation of Natural Language Processing, Taipei, Taiwan, Nov. 2017. 1
- [4] J. Gugenheimer, D. Wolf, G. Haas, S. Krebs, and E. Rukzio. Swivrchair: A motorized swivel chair to nudge users' orientation for 360 degree storytelling in virtual reality. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pp. 1996–2000, 2016. 1
- [5] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The international journal of aviation psychology*, 3(3):203–220, 1993. 1
- [6] J. Kerr, C. M. Kim, K. Goldberg, A. Kanazawa, and M. Tancik. Lrf: Language embedded radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 19729–19739, 2023. 1
- [7] Y.-C. Lin, Y.-J. Chang, H.-N. Hu, H.-T. Cheng, C.-W. Huang, and M. Sun. Tell me where to look: Investigating ways for assisting focus in 360 video. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 2535–2545, 2017. 1
- [8] K. Liu, Y. Chen, J. Liu, X. Zuo, and J. Zhao. Extracting events and their relations from texts: A survey on recent research progress and challenges. *AI Open*, 1:22–39, 2020. 1
- [9] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020. 1
- [10] L. T. Nielsen, M. B. Møller, S. D. Hartmeyer, T. C. Ljung, N. C. Nilsson, R. Nordahl, and S. Serafin. Missing the point: an exploration of how to guide users' attention during cinematic virtual reality. In *Proceedings of the 22nd ACM conference on virtual reality software and technology*, pp. 229–232, 2016. 1
- [11] OpenAI. Chatgpt: Language model, 2023. Accessed: 2024-09-18. 1
- [12] Y.-L. Qiao, A. Gao, Y. Xu, Y. Feng, J.-B. Huang, and M. C. Lin. Dynamic mesh-aware radiance fields. *ICCV*, 2023. 1
- [13] S. Rothe, D. Buschek, and H. Hußmann. Guidance in cinematic virtual reality-taxonomy, research status and challenges. *Multimodal Technologies and Interaction*, 3(1):19, 2019. 1
- [14] T. Schubert, F. Friedmann, and H. Regenbrecht. The experience of presence: Factor analytic insights. *Presence: Teleoperators & Virtual Environments*, 10(3):266–281, 2001. 2
- [15] W. S. Smith and Y. Tadmor. Nonblurred regions show priority for gaze direction over spatial blur. *The Quarterly Journal of Experimental Psychology*, 66(5):927–945, 2013. 1